

Shipping Performance & Profitability Analysis

Business Intelligence Project using Microsoft Power BI

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Tool: Microsoft Power BI

Project Type: End-to-End Business Intelligence Project

1. Project Overview

Background

The international shipping industry operates in a highly competitive environment where profitability depends on transportation costs, freight rates, customer demand, vessel utilization, and operational efficiency.

Management requires timely insights into financial and operational performance in order to make strategic decisions regarding pricing, customer relationships, cost optimization, and route selection.

This project was developed as a complete Business Intelligence solution using Microsoft Power BI to transform raw shipping data into interactive dashboards for executive decision-making.

2. Business Problem

The company lacked a centralized reporting system capable of answering key business questions such as:

- Which shipping routes generate the highest revenue?
- Which customers contribute the largest share of company revenue?
- How are operational costs distributed?
- Which vessels incur the highest operating costs?
- What factors influence shipment delays?
- How would increases in fuel costs affect company profitability?
- Which KPIs should management monitor regularly?

The objective of this project is to provide an interactive reporting solution that enables management to answer these questions in real time.

3. Project Objectives

The primary objectives of this project are:

- Analyze company revenue performance

- Evaluate operational costs
- Measure shipment performance
- Analyze customer profitability
- Compare shipping routes
- Build executive dashboards
- Simulate business scenarios using What-If Parameters
- Support strategic decision making

4. Dataset Description

The project is based on a simulated shipping company dataset containing operational and financial information.

Fact Tables

- Fact_Shipments
- Fact_OperationalCosts

Dimension Tables

- Calendar
- Customers
- Ships
- Routes
- Ports

The data includes information such as:

- Shipment Date
- Revenue
- Freight Rate
- Containers
- Customer
- Route
- Vessel
- Distance
- Fuel Consumption
- Delay
- Fuel Cost
- Crew Cost
- Maintenance Cost
- Insurance Cost
- Canal Fees

5. Data Model

A Star Schema data model was implemented to ensure optimal report performance.

The model consists of:

- Fact tables containing transactional data
- Dimension tables containing descriptive information
- One-to-Many relationships
- Calendar dimension for time intelligence

This modeling approach improves DAX performance and simplifies report development.

6. Data Preparation (Power Query)

Before loading the data into the model, several transformations were performed using Power Query.

These include:

- Data type conversion
- Removing unnecessary columns
- Cleaning inconsistent values
- Creating calculated columns
- Formatting date fields
- Standardizing categorical values
- Preparing data for analysis

7. DAX Measures

More than forty business measures were created throughout the project.

Some of the most important measures include:

- Total Revenue
- Total Profit
- Total Operational Cost
- Total Shipments
- Total Containers
- Profit Margin
- Average Freight Rate
- Average Revenue per Shipment
- Average Revenue per Container
- Average Delay
- Delay Rate
- Average Fuel Consumption
- Average Distance
- Average Transport Duration
- Top Customer Revenue
- Top Route Revenue
- Key Customer Revenue Share
- Projected Fuel Cost

- Projected Operational Cost
- Projected Revenue
- Projected Profit
- Profit Difference %
- Dynamic Titles

8. Dashboards

The report consists of seven interactive dashboards.

Executive Dashboard

Provides a high-level overview of company performance including:

- Revenue
- Profit
- Costs
- Margin
- Shipments
- Revenue Trend
- Customer Segment
- Revenue by Country
- Revenue by Route

Revenue Analysis

Focuses on revenue performance.

Key analyses include:

- Monthly Revenue
- Revenue by Customer
- Revenue by Route
- Revenue by Country
- Customer Segmentation
- Average Revenue per Shipment

Cost Analysis

Analyzes operational expenses.

Includes:

- Fuel Costs
- Maintenance Costs

- Insurance Costs
- Crew Fees
- Cost Structure
- Cost Distribution
- Cost Trends

Operational Performance

Measures shipping efficiency.

Includes:

- Delay Rate
- Average Delay
- Fuel Consumption
- Transport Duration
- Containers
- Shipment Trend

Customer Analysis

Analyzes customer profitability.

Includes:

- Top Customers
- Average Revenue per customer
- Revenue Distribution
- Profit by Customer

Advanced Power BI features:

- Drill-through
- AI Visuals
- Smart Narrative

Route Analysis

Evaluates shipping routes.

Includes:

- Revenue by Route
- Route Performance
- Scatter Analysis
- Decomposition Tree

Forecast

Provides scenario analysis for management.

Includes:

- What-If Parameters
- Fuel Cost Simulation
- Projected Costs
- Projected Profit
- KPI Comparison
- Forecast Trend

9. Advanced Power BI Features

Several advanced Power BI capabilities were implemented throughout the project.

These include:

- Star Schema Modeling
- Power Query Transformations
- DAX Measures
- Dynamic Titles
- What-If Parameters
- Drill-through Pages
- Report Page Tooltips
- AI Visuals
- Interactive Filtering
- Navigation Buttons
- Bookmarks
- Conditional Formatting

10. Business Insights

The dashboards provide valuable insights for management, including:

- Identification of the highest-performing shipping routes
- Recognition of key customers driving revenue
- Visibility into operational cost distribution
- Monitoring shipment delays
- Understanding vessel performance
- Forecasting profitability under changing fuel prices
- Supporting pricing strategy decisions

11. Project Outcomes

The final Power BI solution enables management to:

- Monitor KPIs in real time
- Analyze financial performance
- Track operational efficiency
- Compare customer profitability
- Evaluate shipping routes
- Simulate business scenarios
- Support strategic decision making through interactive dashboards

12. Technologies Used

- Microsoft Power BI Desktop
- Power Query
- DAX
- Microsoft Excel

13. Repository Structure

Shipping-PowerBI-Project

```
|
├── Shipping_Report.pbix
├── Bank_Data.xlsx
├── Documentation.pdf
├── README.md
└── Screenshots
```

14. Conclusion

This project demonstrates an end-to-end Business Intelligence workflow, covering data preparation, data modeling, DAX development, interactive dashboard design, forecasting, and executive reporting.

It highlights practical Power BI skills required for real-world business analytics projects and showcases how data can be transformed into actionable insights for management decision-making.